

L-13

- ① $X_n \rightarrow X$ a.s. $\Leftrightarrow P(\{\omega \mid X_n(\omega) \rightarrow X(\omega)\}) = 1$
- ② $X_n \xrightarrow{P} X \Leftrightarrow \lim_{n \rightarrow \infty} P(|X_n - X| > \varepsilon) = 0 \ \forall \varepsilon > 0.$

Does $E[X_n]$ converge as $n \rightarrow \infty$??

Q) Does convergence of $E[X_n]$ tell us anything?

Example

$$X_n = \begin{cases} n & \text{w.p. } 1/n \\ 0 & \text{o.w.} \end{cases}$$

We have $X_n \geq 0$.

The expectation is $E[X_n] = n \cdot \frac{1}{n} + 0 \cdot (1 - \frac{1}{n}) = 1$ for all $n \geq 1$.

However, $P(X_n \neq 0) = 1/n \downarrow 0$.

This implies that $X_n \xrightarrow{P} 0$, since for any $\varepsilon > 0$, for large enough n , So,
 $P(|X_n - 0| > \varepsilon) = P(X_n = n) = 1/n \rightarrow 0$.

So we have $X_n \xrightarrow{P} 0$ but $E[X_n] = 1 \rightarrow 1$ as $n \rightarrow \infty$.

$$\sum_{n=1}^{\infty} P(X_n \neq 0) = \sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

So if the sequence $(X_n)_{n \geq 1}$ consists of independent random variables, then by the 2nd Borel-Cantelli Lemma, we have $P(X_n \neq 0 \text{ i.o.}) = 1$, which means $P(X_n = n \text{ i.o.}) = 1$.

Definition: We say X_n converges to X in L_1 and write

$$X_n \xrightarrow{L_1} X$$

if $E[|X_n - X|] \rightarrow 0$ as $n \rightarrow \infty$.

Note: If $X_n \xrightarrow{L_1} X$, then $E[X_n] \rightarrow E[X]$ as $n \rightarrow \infty$. But need not be otherwise.

The proof follows from the triangle inequality for expectations:

$$|E[X_n] - E[X]| = |E[X_n - X]| \leq E[|X_n - X|]$$

Since $X_n \xrightarrow{L_1} X$, we can conclude $E[|X_n - X|] \rightarrow 0$ as $n \rightarrow \infty$.

Therefore, $|E[X_n] - E[X]| \rightarrow 0$, which implies $\lim_{n \rightarrow \infty} E[X_n] = E[X]$ using the above result.

The converse is not true.

For our example, we know $E[X_n] = 1$. Let's see if X_n converges in L_1 to the constant random variable $X = 1$.

Q: Does $X_n \xrightarrow{L_1} X = 1$ hold?

$E[|X_n - 1|] = ??$.

Recall

$$X_n = \begin{cases} n & \text{w.p. } 1/n \\ 0 & \text{o.w.} \end{cases}$$

$$\begin{aligned} E[|X_n - 1|] &= |n - 1| \cdot P(X_n = n) + |0 - 1| \cdot P(X_n = 0) \\ &= (n - 1) \cdot \frac{1}{n} + 1 \cdot \left(1 - \frac{1}{n}\right) \\ &= 1 - \frac{1}{n} + 1 - \frac{1}{n} \\ &= 2 \left(1 - \frac{1}{n}\right) \rightarrow 2 \text{ as } n \rightarrow \infty. \end{aligned}$$

Since the limit is not 0, X_n does not converge to $X = 1$ in L_1 .

Theorem: $X_n \xrightarrow{L_1} X \Rightarrow X_n \xrightarrow{P} X$.

But not the other way.

Example: The sequence $X_n = \begin{cases} n & \text{w.p. } 1/n \\ 0 & \text{o.w.} \end{cases}$ shows that convergence in probability does not imply convergence in L_1 . We showed $X_n \xrightarrow{P} 0$, but $E[|X_n - 0|] = E[X_n] = 1$, which does not go to 0.

Proof of Theorem: For any fixed $\varepsilon > 0$, by Markov's inequality:

$$P(|X_n - X| > \varepsilon) \leq \frac{E[|X_n - X|]}{\varepsilon} \rightarrow 0$$

Since $X_n \xrightarrow{L_1} X$, $E[|X_n - X|] \rightarrow 0$. Thus, $P(|X_n - X| > \varepsilon) \rightarrow 0$.

Summary of Convergence Modes:

- ① Almost Sure Convergence: $X_n \rightarrow X$ a.s.
- ② Convergence in Probability: $X_n \xrightarrow{P} X$
- ③ Convergence in L_p : $X_n \xrightarrow{L_p} X$, $p \geq 1$

Relationships: ① $\Rightarrow$ ②, ③ $\Rightarrow$ ②, ② $\nRightarrow$ ①, ② $\nRightarrow$ ③.

Definition: Let $p > 1$ and consider the set of random variables X such that $E[|X|^p] < \infty$. We will write this set as:

$$L_p = \{X : \Omega \rightarrow \mathbb{R} \mid E[|X|^p] < \infty\}.$$

This is called the L_p space.

On this space, we define a notion of convergence:

$$X_n \xrightarrow{L_p} X \iff \lim_{n \rightarrow \infty} E[|X_n - X|^p] = 0,$$

In particular when $p = 1$ (L_1 convergence) and $p = 2$ (L_2 or mean-square convergence). Two special values of p that we will need are $p = 1$ and $p = 2$.

Theorem: For $p \geq 1$, $X_n \xrightarrow{L_p} X \Rightarrow X_n \xrightarrow{P} X$.

Proof: For any $\varepsilon > 0$,

$$\begin{aligned} P(|X_n - X| > \varepsilon) &= P(|X_n - X|^p > \varepsilon^p) \\ &\leq \frac{E[|X_n - X|^p]}{\varepsilon^p} \quad (\text{by Markov's inequality}) \end{aligned}$$

As $n \rightarrow \infty$, the numerator $E[|X_n - X|^p] \rightarrow 0$ by the definition of L_p convergence. Therefore, $P(|X_n - X| > \varepsilon) \rightarrow 0$.

L_2 - Convergence and its relation with WLLN

Recall the Weak Law of Large Numbers (WLLN): If $X_1, X_2, \dots$ are i.i.d. random variables with mean μ and finite variance σ^2 , and $S_n = X_1 + \dots + X_n$, then $S_n/n \xrightarrow{P} \mu$ as $n \rightarrow \infty$.

We can show a stronger result using L_2 convergence. Consider the mean squared error:

$$E \left[\left(\frac{S_n}{n} - \mu \right)^2 \right] = \text{Var} \left(\frac{S_n}{n} \right) = \frac{\text{Var}(S_n)}{n^2} = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

As $n \rightarrow \infty$,

$$E \left[\left(\frac{S_n}{n} - \mu \right)^2 \right] = \frac{\sigma^2}{n} \rightarrow 0.$$

This is precisely the definition of convergence in L_2 .

Therefore, $S_n/n \rightarrow \mu$ in L_2 .